

Endothelial function in healthy 11-year-old children after dietary intervention with onset in infancy: the Special Turku Coronary Risk Factor Inter...

BACKGROUND: Early childhood introduction of nutritional habits aimed at atherosclerosis prevention reduces children's serum total cholesterol concentration, but its effect on vascular endothelial function is unknown. **METHODS AND RESULTS:** Between 1990 and 1992, we randomized healthy 7-month-old infants (n=1062) to intervention (low-saturated-fat diet) and control (unrestricted diet) groups. At the age of 11 years, endothelium-dependent (flow-mediated) and endothelium-independent (nitrate-mediated) vasodilatory responses of the brachial artery were measured with high-resolution ultrasound in 179 intervention and 190 control children. The effect of intervention on endothelial function was significant in boys (P=0.0034) but not in girls (P=0.69). The maximum endothelium-dependent dilation response (mean $\pm$ SD) was 9.62 $\pm$ 3.53% and 8.36 $\pm$ 3.85% in intervention boys and control boys and 8.84 $\pm$ 4.00% and 8.44 $\pm$ 3.60% in intervention girls and control girls, respectively. Intervention had no effect on nitrate-mediated dilation. The difference in endothelial function in boys remained significant after adjustment for current serum total or LDL cholesterol but became nonsignificant after adjustment for mean cholesterol measured under 3 years of age (adjusted means: 9.46% [CI 8.68% to 10.24%] versus 8.54% [CI 7.75% to 9.32%], P=0.11). **CONCLUSIONS:** A low-saturated-fat diet introduced in infancy and maintained during the first decade of life is associated with enhanced endothelial function in boys. The effect is explained in part by the diet-induced reduction in serum cholesterol concentration.